CS 476 - Project Proposal
Korbin Wyssen
Nicholas Fuzesy
Reetinder Singh
Aarav Gupta
Smitsinh Nileshsinh Thakor

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Dr. Samira Sadaoui
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SCLibrary

Sharing is Caring Library

Description - Korbin

SCLibrary is a physical media database where users can create a personal library under their account for all the physical media they own (books, movies, music, video games, audiobooks, etc). They can keep track of the type (such as disc format for movies, and console format for video games), create collections out of their library, and assign tags to individual items. From this, users can see what media other friends have available, and be able to request to borrow from them. Users can assign an optional whitelist or blacklist of media they do or don't want to have available to others for lending.

A user can request to borrow an item in another user's library only if that library has been shared with them. From here, we define the user that wants to borrow an item as the "**borrower**", and the user who is lending out the item as the "**lender**." When a user requests to borrow an item, the lender will be notified in the web-app that the borrower requested for an item to borrow. From there the lender can view the request, and choose to either accept or deny it (the borrower will be notified of the response). If the request is accepted, the lender can set a return date (agreed upon by the lender and the borrower prior through external communication). After the request has been accepted, when the items is being viewed, its status will be shown as (for example):

- "Being borrowed" to other users
- "Being borrowed by <username>" to the lender
- "You're borrowing this item" to the borrower

Once the borrower returns the item back to the lender, the lender can set the status of the item back to available. If the status is not reset past the due date, both the lender and the borrower are notified of the item's overdue return.

Lender Role - Aarav & Reetinder

Functional Requirements

The lender must have the ability to:

- create a personal library
- populate their personal library with physically owned items
- add/remove materials from their library
- mark an item as available or unavailable for borrowing
- accept or decline borrow requests
- set/confirm a return date
- see which items they own are currently being borrowed, and by who
- mark an item as returned

Quality Requirements

- The system should notify the lender of new borrow requests and overdue items

The lender should have the ability to:

- rate and/or review borrowers after an item is returned
- keep a history of previously lent materials
- add notes about the condition of an item before/after lending
- whitelist or blacklist certain materials from the shared library
- organize the personal library using collections/tags

Borrower Role - Nicholas

Functional Requirements

The borrower must have the ability to:

- view a lender's personal library
- see items that are available to be borrowed
- make a request for an item from a lender
- suggest a return date in the request

Quality Requirements

- The system should have a karma score, rewarding borrowers for returning items on time and in good condition
- The system should have a system to notify borrowers, reminding them to return an item

The borrower should have the ability to:

- rate and review borrowed items
- customize their personal library of borrowed items
- create a wishlist for items